

# Yifan Zhang

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## RESEARCH INTERESTS

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My research focuses on AI for Code and Data-Centric AI for Systems. I aim to develop reliable systems by integrating fine-tuning and reasoning in LLMs with neuro-symbolic code representations, leveraging both static and runtime data to enhance robustness and efficiency in agentic workflows. I am also broadly interested in AI applications beyond software engineering, including databases, cyber-physical systems, and medical imaging.

Last updated: 07/16/2026. Legally authorized to work in the U.S. without employer sponsorship.

## EDUCATION

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### Vanderbilt University

*Ph.D. in Computer Science, Specialize in AI for Software Engineering*

*Nashville, TN, USA*

*May. 2022 – May. 2026*

- Advised by Prof. Kevin J. Leach & Prof. Yu Huang

### Georgia Institute of Technology

*M.Sc. in Computer Science, Specialize in Interactive Intelligence/Artificial Intelligence*

*Atlanta, GA, USA*

*Aug. 2022 – Aug. 2026*

- Studied courses in RL and program analysis

### China University of Petroleum

*M.Eng. in Petroleum Engineering, Specialize in Industrial & System Engineering*

*Beijing, CN*

*Sep. 2012 – Jun. 2019*

- Double majored in English (TEM-8 holder) & minored in British Parliamentary debates

### University of Calgary

*Undergraduate Exchange Program in Petroleum Engineering*

*Calgary, AB, CA*

*Dec. 2015 – Jun. 2016*

- Fully-funded by Chinese national fellowship for overseas studies

## PUBLICATIONS (\*ALPHABETICAL; †MENTOR)

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### Papers In Submitting & Under Review

- [R1] **Yifan Zhang\***, Xiaohan Wang\*, Yueke Zhang, Yu Huang, Kevin Leach: *Constraint-Guided Multi-Agent Decompile for Executable Binary Recovery*. Under Review. [\[arXiv\]](#)
- [R2] **Yifan Zhang**, Michael Sandborn, Yu Huang, Kevin Leach: *STAR: Structure-Transformed Agentic Reasoning for End-to-End Software Engineering Tasks*. In Submitting.
- [R3] Zihan Fang, **Yifan Zhang†**, Yueke Zhang, Kevin Leach, Yu Huang: *DPO-F+: Aligning Code Repair Feedback with Developer Preferences*. Under Review. [\[arXiv\]](#)
- [R4] Weiheng Qiu, **Yifan Zhang**, Kevin Leach: *G-BTX: Scalable Semantic Recovery in Executables via Graph-Neural Dependency Modeling*. Under Review. [\[arXiv\]](#)
- [R5] Yueke Zhang, **Yifan Zhang†**, Kevin Leach, Yu Huang: *CodeGrad: Integrating Multi-Step Verification with Gradient-Based LLM Refinement*. Under Review. [\[arXiv\]](#)
- [R6] Suad Mohamed, Zachary Karas, **Yifan Zhang†**, Aakash Bansal, Collin McMillan, Yu Huang: *The Potential of Human Attention Patterns to Predict Code Summary Quality*. In submitting.
- [R7] Yueke Zhang, **Yifan Zhang**, Zihan Fang, Kevin Leach, Yu Huang: *Stealthy Paper*. In submitting.

### Refereed Long Journal & Conference Papers

- [P1] Yueke Zhang, **Yifan Zhang**, Wisniewski Pamela, Fengwei Zhang, Kevin Leach, Yu Huang: *Understanding and Detecting GitHub Impersonation through Automated Authorship Attribution*. The ACM Transactions on Software Engineering and Methodology (**TOSEM**), 2026. SJR Q1. [\[Paper\]](#)
- [P2] **Yifan Zhang**, Chen Huang, Yueke Zhang, Jiahao Zhang, Toby Jia-Jun Li, Collin McMillan, Kevin Leach, Yu Huang: *EyeMulator: Improving Code Language Models by Mimicking Human Visual Attention*. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (**ACL'26**), July 2-7, 2026. CORE A\*. **ACL'26 Oral Presentation**. [\[Paper\]](#) [\[arXiv\]](#) [\[artifact\]](#) [\[poster\]](#) [\[slides\]](#) [\[video\]](#)

- [P3] **Yifan Zhang**, Jieyu Li, Kexin Pei, Kevin Leach, Yu Huang: *SynthFix: A Hybrid Neuro-Symbolic Framework for Code Vulnerability Repair*. In Findings of the 63rd Annual Meeting of the Association for Computational Linguistics (**ACL’26 Findings**), July 2-7, 2026. [\[Paper\]](#) [\[arXiv\]](#) [\[artifact\]](#) [\[poster\]](#) [\[slides\]](#) [\[video\]](#).
- [P4] Yueke Zhang\*, **Yifan Zhang**\*†, Zihan Fang, Greg Trafton, Daniel Levin, Kevin Leach, Yu Huang: *ScanCoder: Leveraging Human Attention Patterns to Enhance LLMs for Code*, In Proceedings of the ACM International Conference on the Foundations of Software Engineering (**FSE’26**), July 5-9, 2026. CORE A\*. [\[Paper\]](#) [\[slides\]](#)
- [P5] Jiahao Zhang, **Yifan Zhang**†, Kevin Leach, Yu Huang: *EyeLayer: Integrating Human Attention Patterns into LLM-Based Code Summarization*. In Proceedings of the 34th IEEE/ACM International Conference on Program Comprehension (**ICPC’26**). April 12-13, 2026. CORE A. [\[arXiv\]](#) [\[video\]](#)
- [P6] **Yifan Zhang**, Giridhar Ganapavarapu, Srideepika Jayaraman, Bhavna Agrawal, Dhaval Patel, Achille Fokoue: *SPIRAL: Symbolic LLM Planning via Grounded and Reflective Search*. In Proceedings of the Fortieth AAAI Conference on Artificial Intelligence (**AAAI’26**), January 20-January 27, 2026. CORE A\*. [\[Paper\]](#) [\[arXiv\]](#) [\[artifact\]](#) [\[poster\]](#) [\[slides\]](#) [\[video\]](#)
- [P7] **Yifan Zhang**, Chen Huang, Yueke Zhang, Huajie (Jay) Shao, Kevin Leach, Yu Huang: *Pre-Training Representations of Binary Code Using Contrastive Learning*. The Transactions on Machine Learning Research (**TMLR**), 2025. [\[Paper\]](#) [\[arXiv\]](#) [\[artifact\]](#) [\[slides\]](#) [\[video\]](#)
- [P8] Xinwei Yang, Zhaofeng Liu, Chen Huang, Jiashuai Zhang, Tong Zhang, **Yifan Zhang**, Wenqiang Lei: *ELABORATION: A Comprehensive Benchmark on Human-LLM Competitive Programming*. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (**ACL’25**), July 27-August 1, 2025. CORE A\*. [\[Paper\]](#) [\[arXiv\]](#) [\[artifact\]](#)
- [P9] Jiliang (Eric) Li\*, **Yifan Zhang**\*†, Yu Huang, Kevin Leach: *MalMixer: Few-Shot Malware Classification with Retrieval-Augmented Semi-Supervised Learning*. In Proceedings of the 10th IEEE European Symposium on Security and Privacy (**EuroS&P’25**), June 30-July 4, 2025. CORE A. [\[Paper\]](#) [\[arXiv\]](#) [\[artifact\]](#) [\[poster\]](#) [\[slides\]](#) [\[video\]](#)
- [P10] **Yifan Zhang**, Xue (Sharon) Yang: *CONSTRUCTA: Automating Commercial Construction Schedules in Fabrication Facilities with Large Language Models*. In Proceedings of the 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (**NAACL’25**). April 29–May 4, 2025. Industry Track Paper. CORE A. [\[Paper\]](#) [\[arXiv\]](#) [\[poster\]](#) [\[slides\]](#) [\[video\]](#)
- [P11] Yichang He, **Yifan Zhang**, Yunpeng Fan, U-Xuan Tan: *Real-Time Vibration Compensation with Long Short-Term Memory Recurrent Neural Network and Adaptive Filter*. The IEEE/ASME Transactions on Mechatronics (**TMech**), 2024. SJR Q1. [\[Paper\]](#).
- [P12] Zachary Karas, Aakash Bansal, **Yifan Zhang**, Toby Jia-Jun Li, Collin McMillan, Yu Huang: *A Tale of Two Comprehensions? Studying Human Attention during Code Summarization*. The ACM Transactions on Software Engineering and Methodology (**TOSEM**), 2024. SJR Q1. **ICSE’25 Journal-First Paper**. [\[Paper\]](#) [\[artifact\]](#) [\[slides\]](#)
- [P13] **Yifan Zhang**, Jiliang (Eric) Li, Zachary Karas, Aakash Bansal, Toby Jia-Jun Li, Collin McMillan, Kevin Leach, Yu Huang: *EyeTrans: Merging Human and Machine Attention for Neural Code Summarization*. In Proceedings of the ACM International Conference on the Foundations of Software Engineering (**FSE’24**), July 15-19, 2024. CORE A\*. **The C. F. Chen Best Paper Award**. [\[Paper\]](#) [\[arXiv\]](#) [\[artifact\]](#) [\[poster\]](#) [\[slides\]](#)
- [P14] Chen Huang, Haoyang Li, **Yifan Zhang**, Wenqiang Lei, Jiancheng Lv: *Cross-Space Adaptive Filter: Integrating Graph Topology and Node Attributes for Alleviating the Over-smoothing Problem*. In Proceedings of the ACM Web Conference (**WWW’24**), May 13-17, 2024. CORE A\*. [\[Paper\]](#) [\[arXiv\]](#) [\[poster\]](#) [\[slides\]](#) [\[video\]](#)
- [P15] Haoyu Dong, **Yifan Zhang**, Hanxue Gu, Nicholas Konz, Maciej Mazurowski: *SWSSL: Sliding-Window based Self-Supervised Learning Framework for Anomaly Detection*. The IEEE Transactions on Medical Imaging (**TMI**), 2023. SJR Q1. [\[Paper\]](#) [\[artifact\]](#)
- [P16] Jing Li, Xiangfang Li, Keliu Wu, Dong Feng, Tao Zhang, **Yifan Zhang**: *Thickness and Stability of Water Film Confined inside Nanoslits and Nanocapillaries of Shale and Clay*. The International Journal of Coal Geology (**IJCG**), 2017. SJR Q1. [\[Paper\]](#)

## Refereed Short Conference Papers

- [S1] **Yifan Zhang**, Chen Huang, Zachary Karas, Thuy Dung Nguyen, Kevin Leach, Yu Huang: *Enhancing Code LLM Training with Programmer Attention*. The 33rd ACM International Conference on the Foundations of Software Engineering (**FSE’25 Companion**). June 23-27, 2025. IVR Track Paper. [\[Paper\]](#) [\[arXiv\]](#) [\[slides\]](#)

- [S2] Manish Acharya\*, **Yifan Zhang**\*<sup>†</sup>, Kevin Leach, Yu Huang: *Optimizing Code Runtime Performance through Context-Aware Retrieval-Augmented Generation*. In Proceedings of the 33rd IEEE/ACM International Conference on Program Comprehension (**ICPC'25**). April 27-28, 2025. ERA Track Paper. CORE A. [\[Paper\]](#) [\[arXiv\]](#) [\[artifact\]](#) [\[slides\]](#)
- [S3] Jiliang (Eric) Li, **Yifan Zhang**<sup>†</sup>, Zachary Karas, Collin McMillan, Kevin Leach, Yu Huang : *Do Machines and Humans Focus on Similar Code? Exploring Explainability of Large Language Models in Code Summarization*. In Proceedings of 32nd IEEE/ACM International Conference on Program Comprehension (**ICPC'24**), April 15-16, 2024. RENE Track Paper. CORE A. [\[Paper\]](#) [\[arXiv\]](#) [\[slides\]](#)
- [S4] Aakash Bansal, Chia-Yi Su, Zachary Karas, **Yifan Zhang**, Yu Huang, Toby Jia-Jun Li, Collin McMillan: *Modeling Programmer Attention as Scanpath Prediction*. In Proceedings of the 38th IEEE/ACM International Conference on Automated Software Engineering (**ASE'23**), September 11-15, 2023. NIER Track Paper. CORE A\*. [\[Paper\]](#) [\[arXiv\]](#)

## Refereed Extended Abstract & Position Papers

- [A1] **Yifan Zhang**, Kevin Leach: *Training Large Language Models to Comprehend LLVM IR via Feedback-Driven Optimization*. In Companion Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering (**FSE'25 Companion**). June 23-27, 2025. HumanAISE'25 Position Paper. [\[Paper\]](#) [\[slides\]](#)
- [A2] **Yifan Zhang**, Kevin Leach: *Leveraging Human Insights for Enhanced LLM-based Code Repair*. In Companion Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering (**FSE'25 Companion**). June 23-27, 2025. HumanAISE'25 Position Paper. [\[Paper\]](#) [\[slides\]](#)
- [A3] **Yifan Zhang**: *Leveraging Artificial Intelligence on Binary Code Comprehension*. In Proceedings of the 37th IEEE/ACM International Conference on Automated Software Engineering (**ASE'22**), October 10-14, 2022. Doctoral Symposium Paper. CORE A\*. [\[Paper\]](#) [\[arXiv\]](#) [\[poster\]](#) [\[slides\]](#)

## Refereed Workshop & Symposium Papers

- [W1] Manish Acharya, Zhenyu Liao, Yuke Zhang Kevin Leach, Yu Huang, **Yifan Zhang**: *VERITAS: Verifier-Guided Proof Search for Zero-Shot Formal Theorem Proving*. Presented at the 43rd International Conference on Machine Learning Workshops (ICMLW'26). July 11, 2026.
- [W2] **Yifan Zhang**\*, Michael Sandborn\*, Stefan Larson, Yu Huang, Kevin Leach: *K-ASTRO: Structure-Aware Adaptation of LLMs for Code Vulnerability Detection*. Presented at the Hot Topics in the Science of Security Symposium 2025 (HotSoS'25). April 2-4, 2025. WIP Paper. [\[webpage\]](#) [\[WIP\]](#) [\[slides\]](#)
- [W3] **Yifan Zhang**, Junwen Yang, Haoyu Dong, Qingchen Wang, Huajie (Jay) Shao, Kevin Leach, Yu Huang: *ASTRO: An AST-Assisted Approach for Generalizable Neural Clone Detection*. Presented at the 45th International Conference on Software Engineering Workshops (ICSEW'23), May 14th, 2023. Short Paper. Extended to be in HotSoS'25. [\[WIP\]](#) [\[slides\]](#)
- [W4] **Yifan Zhang**\*, Haoyu Dong\*, Nicholas Konz, Hanxue Gu, Maciej Mazurowski: *Lightweight Transformer Backbone for Medical Object Detection*. In Workshop Proceedings of the 25th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAIW'22), September 22nd, 2022. [\[Paper\]](#) [\[WIP\]](#) [\[poster\]](#) [\[slides\]](#) [\[video\]](#)

## INDUSTRY EXPERIENCE

### Salesforce AI Research

Research Scientist, Supervisor: Dr. Zeyuan Chen & Dr. Ran Xu

Jan. 2026 – Present

Palo Alto, CA, USA

- **Enterprise Computer-Use Agent**: Architecting an autonomous system for complex CRM workflows. Planned research leverages an Action-as-Code paradigm to replace fragile GUI interactions with robust programmatic execution, ensuring long-horizon consistency across fragmented enterprise environments.

### Amazon Web Services

Applied Scientist Intern, Supervisor: Dr. Brandon Paulsen & Dr. Joey Dodds

Sep. 2025 – Dec. 2025

Arlington, VA, USA

- **Cloud Infrastructure Simulation**: Developed CloudTwin, an agentic framework using a Operation-as-State paradigm to maintain stateful digital twins of AWS infrastructure. Designed HCM to scale reasoning across 200+ operations while preserving ground-truth consistency for formal verification. Integrated LLMs with AWS schema validation to enable end-to-end IaC testing without physical resource provisioning.

## IBM T.J. Watson Research

AI Research Intern, Supervisor: Dr. Dhaval Patel & Dr. Achille Fokoue

May. 2025 – Aug. 2025

Yorktown Heights, NY, USA

- **Symbolic Reasoning Framework:** Architected a self-improving planning framework with the IBM Watson AI pillar, where a parallelized Monte Carlo Tree Search was guided by a symbolic value function derived from a dynamically learned rule base. The system used the LLM as an inductive learner to discover and refine strategic heuristics from successful plans, enabling efficient, stateful reasoning on complex agentic tasks without costly simulations or model fine-tuning.

## Google Research

Student Researcher, Supervisor: Dr. Alekh Agarwal & Dr. Mircea Trofin

Jan. 2025 – May. 2025

Sunnyvale, CA, USA

- **LLVM IR Benchmark Automation:** Designed an automated pipeline with Google CORE/Google Research, using LLMs to generate C++ code that simulates LLVM IR profiles, enabling benchmarking and performance analysis of production code. The system supported AI models in learning compiler behavior and optimization strategies across diverse hardware platforms, with safeguards in place to preserve IP integrity.

## Intel Corporation

GenAI Research Intern, Supervisor: Dr. Jun Li & Dr. Xue (Sharon) Yang

Jun. 2024 – Dec. 2024

Santa Clara, CA, USA

- **Construction Schedule Automation:** Developed and implemented a LLM-based system with Intel Incubator, to automate and enhance construction scheduling for Intel's semiconductor fabrication projects. Integrated rule-based knowledge construction and in-context learning to predict and adjust schedules, leading to improved 2.8x efficiency compared with GPT-4o and reduced manual intervention.

## TikTok/ByteDance

Research Scientist Intern, Supervisor: Dr. Zhibing Zhao & Dr. Tieying Zhang

May. 2023 – Aug. 2023

San Jose, CA, USA

- **SQL Hint Recommendation:** Enhanced the SQL Pipeline for the ByteBrain team by developing a hint recommendation system based on representation learning. Leveraged two Transformer models in a pipeline and pretrained a generalizable model for reranking SQL plans across diverse table spaces and schemas.
- **Research Collaboration:** Participated in group research activities, including giving a technical talk at TikTok InfraLab and working with other team members on LLM literature review and conference paper review.

## JD.com

Machine Learning Engineer, Supervisors: Dr. Hu Wang & Dr. Chen Huang

Dec. 2018 – Mar. 2021

Beijing, CN

- **Action Model:** Built Bi-GRU and DeepFM models on user behavior features to predict the credit use rate and overall profit of every user in cash loan and consumer debt. The model can propose decisions to increase their credit limit for maximizing income, and achieved 21.4% overall profit increase.
- **Credit Score Propagation:** Built a heterogeneous graph on different types of user connections, and applied GNN models to propagate the credit score and improve risk prediction. The model can improve the overall accuracy of the XGB model by 5% in user classification.
- **Privacy-Preserving Collaboration:** Invented one kind of GAN-styled model using differential privacy to improve the efficiency and security of federated learning. Granted 5 CN patents based on the research outputs. One of the patents was awarded as 1st Runner-up in the 3rd JD Discovery Cup Patent Competition.

## ACADEMIC EXPERIENCE

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### Duke University

Research Associate, Advisor: Prof. Maciej A. Mazurowski

Jul. 2021 – Jun. 2022

Durham, NC, USA

- **Medical Object Detection:** Designed a feature interpolation pipeline for injecting tumors into healthy images as an augmented dataset, and conjuncted a ViT on the outputs of a ResNet as inputs to a FPN in Faster R-CNN for tumor detection. The model mitigates the data-hungry problem of attention and achieves 13.1% improvement in AP50 for detecting tumors.

### University of Hong Kong

Senior Research Assistant, Advisor: Prof. Qingchen Wang

Mar. 2021 – Sep. 2021

Hong Kong SAR

- **Financial Decision Making:** Built an entire intelligent debt collection system using data-driven deep reinforcement learning models. The model utilizes Transformer as the feature extractor and attaches a offline policy gradient model trained on the embedded sequential-aware hidden features to propose long-term decisions.

### Program Committee Member & Conference Reviewer

- 2026 Annual Conference on Neural Information Processing Systems (NeurIPS'26)
- 2026 Annual Meeting of the Association for Computational Linguistics Rolling Review (ACL ARR'26)
- 2026 International Workshop on Large Language Models for Code (LLM4Code@ICSE'26)
- 2026 AAAI Conference on Artificial Intelligence (AAAI'26)
- 2025 Annual Meeting of the Association for Computational Linguistics (ACL'25) Industry Track
- 2025 Annual Conference on Neural Information Processing Systems (NeurIPS'25) Datasets & Benchmarks Track
- 2025 International Conference on Machine Learning (ICML'25)
- 2025 International Conference on Learning Representations (ICLR'25)
- 2024 Annual Conference on Neural Information Processing Systems (NeurIPS'24) Datasets & Benchmarks Track
- 2024 Annual Conference on Neural Information Processing Systems (NeurIPS'24) [\[webpage\]](#)
- 2024 ACM Conference on Computer and Communications Security (CCS'24) Artifact Evaluation Track [\[webpage\]](#)
- 2024 International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI'24)
- 2024 Data-Centric Machine Learning Research Workshop at ICLR (DMLR@ICLR'24) with **Exceptional Reviewer Award** (Top 9.43% venue-wide) [\[webpage\]](#)
- 2024 AAAI Conference on Artificial Intelligence (AAAI'24)
- 2023 Annual Conference on Machine Learning and Systems (MLSys'23) Artifact Evaluation Track
- 2023 International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI'23)
- 2023 AAAI Workshop on DL-Hardware Co-Design for AI Acceleration (DCAA@AAAI'23)
- 2023 International Conference on Mining Software Repositories (MSR'23)
- 2023 AAAI Conference on Artificial Intelligence (AAAI'23)
- 2022 MICCAI Workshop on Cancer Prevention through Early Detection (CaPTion@MICCAI'22)

### External Conference Reviewer

- 2025 The annual IEEE International Conference on Data Engineering (ICDE'25) Industry Track
- 2025 International Conference on Extending Database Technology (EDBT'25) Industry Track
- 2025 Network and Distributed System Security (NDSS'25)
- 2024 International Conference on Very Large Data Bases (VLDB'24) Scalable Data Science Track
- 2023 USENIX Security Symposium (USENIX Security'23)
- 2022 IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR'22)

## RESEARCH AND INSTRUCTIONAL MENTORSHIP

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### Research Mentorship Experience

- Zichen (Ryan) Zhu, Bachelor of Applied Math & CS at Vanderbilt University. From 2025 Spring to Present. Research Topic: Dynamic Sparsity in LLM Compiler Optimization.
- Suad Mohamed, Bachelor of Psychology & CS at Belmont University. *Former Software Engineer Intern at Microsoft*. Co-Advised with Mr. Zachary Karas. From 2024 Spring to Present. Research Topic: Eye-Tracking for Human-Centered AI in Code Summarization. First Placement: Software Engineer at Microsoft.
- Manish Acharya, Bachelor of Math & CS at Vanderbilt University. *Vanderbilt Chancellor's Scholar*. From 2024 Spring to Present. Research Topic: Software Patch In-Context Learning.
- Jieyu Li, Bachelor of EEE at Shanghai Jiaotong University (SJTU), Master student of ECE at Vanderbilt University. From 2023 Fall to Present. Research Topic: AI Agent for Automated Program Repair. First Placement: Ph.D. in ECE at University of Waterloo. Advisor: Prof. Weiyi (Ian) Shang.
- Luka Mushkudiani, Bachelor of Math & CS at Vanderbilt University. *Former Intern at Meta's PyTorch Compiler Team*. From 2023 Fall to Present. Research Topic: Neural Type Systems for Python/JavaScript
- Jiliang (Eric) Li, Bachelor of Math & CS at Vanderbilt University. From 2022 Summer to Present. Research Topic: Few-Shot Malware Classification. First Placement: Master of Science in Computer Science (MScS) at Stanford University.

### Teaching and Instructional Experience

- Graduate Teaching Assistant for CS3892: Projects in Computing for Sustainability at Vanderbilt University, *Undergraduate Capstone Project*, 2024 Spring. Instructor: Prof. Douglas H. Fisher.
- Graduate Teaching Assistant for CS3276/CS5276: Compiler Design at Vanderbilt University, 2023 Fall. Instructor: Prof. Kevin J. Leach.



- Community Teaching Assistant for Data Structure and Algorithm Training Camp (Part I and Part II) at Tsinghua University (MOOC), 2019 Spring. Instructor: Prof. Junhui Deng. [\[certificate\]](#)
- Online Course Mentor for Data Analysis Nanodegree at Udacity China, 2017 Spring. Instructors: Dr. Josh Magee, Ria Cheruvu & Matt Maybeno. **Outstanding Mentor Award** (Top 5% world-wide) [\[certificate\]](#)

## FUNDS AND AWARDS

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### Scholarships and Fellowships

- 2025 Student Travel Support from National Science Foundation (NSF) for FSE'25/ISSTA'25 (\$3000 Funding)
- 2025 The C.F. Chen Best Paper Award from Vanderbilt University (Top 1 university-wide) (\$5000 Scholarship) [\[certificate\]](#) [\[webpage\]](#)
- 2021 Research Fellowship from National Institutes of Health (NIH) (\$36000/year)
- 2017 Roberto Roca Education Scholarship (Top 10 nationwide)
- 2015 Chinese National Fellowship for Overseas Studies (Full tuition fees & CA\$6000/4 months)
- 2015 & 2016 & 2017 & 2018 First-class Scholarship at China University of Petroleum (CUP)
- 2014 Schlumberger Engineering Scholarship (Top 8 university-wide)
- 2013 Chinese National Scholarship for Outstanding Merits (Top 0.2% nationwide)

### Honors and Awards

- 2026 Featured Research Recognition from Vanderbilt Institute Newsletter for ACL'26 Oral Presentation (Top 4% venue-wide) [\[webpage\]](#)
- 2026 Travel Award from ACM SIGSOFT CAPS for FSE'25/ISSTA'25 Conference Attendance
- 2025 Travel Award from ACM SIGSOFT CAPS for FSE'25/ISSTA'25 Conference Attendance
- 2020 First Runner-up in the 3rd JD.com Discovery Cup Patent Competition (Top 2 company-wide)
- 2020 Silver Medal Award of Distinguished Technical Recruiter at JD.com (Top 5% company-wide)
- 2020 Bronze Medal Award of Certified Technical Instructor at JD.com
- 2019 Beijing Outstanding Graduate Award (Top 0.1% nationwide)
- 2016 *Summa Cum Laude* for the Best Undergraduate Students at CUP (Top 1% university-wide)
- 2015 & 2017 Meritorious Winner of American Mathematical Contest in Modeling (Top 5% worldwide)
- 2013 & 2015 Third Prizes of Chinese National Petroleum Engineering Design Competition (Top 5% nationwide)